

Applied Farm Management

Science

May, 2026

Applied Farm Management (A08681)

Short Title: Applied Farm Management
Faculty Science and Computing
Department Science
Cluster Agriculture
Author SWALSH
Credits 5

Level: Advanced

Description of Module / Aims

This module is designed to develop the students' understanding of important concepts and applications of soil science, grassland management, animal breeding and reproduction, nutritional strategies and economic parameters. This will be achieved through the holistic evaluation of a farm enterprise and its potential future developments. To enhance the students' learning, they will be exposed to all current research developments.

Programmes

AGRI-0108 BSc (Hons) in Agricultural Science (WD_SAGRI_B)

4 / 7 / M

Indicative Content

- Soil quality and fertility and their effects on feed, forage and cereal production
- Grass budgeting, grassland management and associated software applications
- Animal breeding and reproductive management strategies
- Nutritional management strategies and associated software applications
- Agri-business planning incorporating e-profit monitors, cost control planners, financial planning

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Critique current and existing research in soil science.
2. Evaluate current and evolving grassland management strategies and technologies.
3. Assess animal breeding programs and efficiencies in different production systems.
4. Design optimum fertility management protocols.
5. Appraise and assess current animal feed formulation software.
6. Develop a sustainable farm plan based on using data from a monitor farm.
7. Evaluate their own learning progress.
8. Produce an effective presentation.

Learning and Teaching Methods

- Lectures.
- Guest industry speakers.
- Farm visits.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Presentation</i>	20%	8
<i>Project</i>	70%	1,2,3,4,5,6
<i>Assignment</i>	10%	7

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter.

40%-49%: Will be able to show a good knowledge of the key learning outcomes including soil and grassland management, animal breeding and reproduction, as well as ability to predict nutritional requirements of the animal given a range of genetic, management and environmental factors.

50%-59%: As well as the above, the student will be able to demonstrate a good understanding of the learning outcomes, as well as being able to integrate knowledge assimilated in other modules of the course.

60%-69%: In addition, the student will demonstrate more detailed knowledge of all the topics, including how to critically assess performance indicators of an enterprise, make recommendations for improvements and justify these recommendations.

70%-100%: The student will be able to demonstrate a comprehensive knowledge and understanding of all learning outcomes, including the ability to fully integrate the various components of the course.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- McDonald, P., R.A. Edwards, J.F. Greenhalgh and C.A. Morgan. *Animal Nutrition*. UK: Pearson, 2002.
- O'Dwyer, J. *Farm Management*. Ireland: Teagasc, 2012.
- O'Dwyer, J., J. Maher, S. Norris, F. Phelan and K. Connolly. *Farm Business Planning*. Ireland: Teagasc, 2013.
- Womack, J. *Bovine Genomics*. England: John Wiley & Sons, 2012.

Supplementary Material

- Bedford, M. and G. Partridge. *Enzymes in Farm Animal Nutrition*. UK: CABI Publishing, 2001.
- Givens, D.J., E.J. Owens and H.M. Omed. *Forage Evaluation in Ruminant Nutrition*. UK: CABI Publishing, 2000.
- Whitehead, D.C. *Nutrient Elements in Grassland : Soil-Plant-Animal Relationships*. UK: CABI Publishing, 2000.

Requested Resources

- Room Type: Computer Lab
- Room Type: Biology Lab